CommerceFlow Physical ERD Specification

Version 1.0 | 29 October 2026 | DESIGN DRAFT

Business Tables

users(id UUID PK, email unique case-insensitive, password_hash, full_name, account_status, timestamps); roles(id SMALLINT PK, name UNIQUE); user_roles(user_id, role_id composite PK); categories(id UUID PK, name, slug UNIQUE); products(id UUID PK, category_id FK, name, slug UNIQUE, description, price NUMERIC(12,2), stock_quantity INTEGER, active BOOLEAN, version BIGINT, timestamps); carts(id UUID PK, user_id UNIQUE FK); cart_items(id UUID PK, cart_id FK, product_id FK, quantity INTEGER, UNIQUE(cart_id,product_id)); orders(id UUID PK, user_id FK, order_status, payment_status, total_amount NUMERIC(12,2), created_at); order_items(id UUID PK, order_id FK, product_id FK, product_name, unit_price NUMERIC(12,2), quantity, line_total NUMERIC(12,2)); order_addresses(id UUID PK, order_id UNIQUE FK, recipient_name, address_line1, city, state, postal_code, country_code).

Integrity Rules

Price and stock must be nonnegative; cart/order quantities must be positive; order item names and prices are immutable snapshots; order addresses are snapshots; products referenced by orders cannot be hard-deleted; user email unique via normalized index; cart item pair unique.

Indexes

Index foreign keys for products.category_id, orders.user_id, cart_items.cart_id, order_items.order_id; use product active/category/search indexes based on measured query patterns.

Infrastructure Tables

Spring Session JDBC persistence and checkout idempotency require separate infrastructure tables, not counted among ten business tables. Define their migrations and lifecycle before implementation.

Open Design Questions

Final order cancellation/stock restoration, checkout key retention, payment simulation failure semantics, CSRF transport, physical DDL details and deployment cookie policy require final implementation-level review.